Assignment :- 4

Name: Kaushal kumar Raj
PRN: 240205241023
Branch: B.Tech CSE with AI & ML
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① What is virtual memory? why is it needed? Explain its benefits.

→ Virtual memory is a technique where secondary storage (disk) is used to extend RAM, giving the illusion of a larger memory.

Need:
- Programs exceed physical memory
- Efficient memory use
- supports multitasking.

Benefits:
- Run large programs
- Better memory utilization
- Process isolation
- More processes can run simultaneously.

② Explain the concepts of Demand paging. Describe the steps involved in handling a page fault.

→ Damand paging : Pages are loaded only when required.

Page fault Handling:

1) Page not found in memory
2) Trap to OS
3) check validity
4) Locate page on disk
5) find free frame / replace page
6) load page
7) update page table
8) Restart instruction.

③ What is a page Replacement Algorithm? Name any three algorithms.

→ It decides which page to remove when memory is full.

Examples:

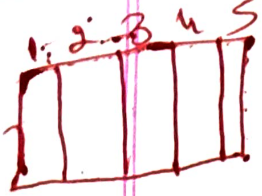
- FIFO
- LRU
- optimal

(4) Given reference string : 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2.
Calculate page faults for FIFO with 4 frames.

⇒

step	page	frame after operation	fault / Hit
1	7	7 - - -	Fault(1)
2	0	7 0 - -	Fault(2)
3	1	7 0 1 -	Fault (3)
4	2	7 0 1 2	Fault(4)
5	0	7 0 1 2	Hit
6	3	3 0 1 2 (7 Replaced)	Fault(5)
7	0	3 0 1 2	Hit
8	4	3 4 1 2 (0 Replaced)	Fault(6)
9	2	3 4 1 2	Hit
10	3	3 4 1 2	Hit
11	0	3 4 0 2 (1 Replaced)	Fault(7)
12	3	3 4 0 2	Hit
13	2	3 4 0 2	Hit

∴ Total faults Page = 7



⑤ for the same reference string (7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2). calculate page fault for optimal with 4 frames.

⇒

Step	Page	Frames after operation	Fault / Hit
1	7	7 - - -	fault(1)
2	0	7 0 - -	fault(2)
3	1	7 0 1 -	fault(3)
4	2	7 0 1 2	fault(4)
5	0	7 0 1 2	Hit
6	3	3 0 1 2 (7 replaced)	fault(5)
7	0	3 0 1 2	Hit
8	4	3 0 4 2 (1 replaced)	fault(6)
9	2	3 0 4 2	Hit
10	3	3 0 4 2	Hit
11	0	3 0 4 2	Hit
12	3	3 0 4 2	Hit
13	2	3 0 4 2	Hit

∴ Total page faults = 6

⑥ What is Belady's Anomaly ? Illustrate with an example of the FIFO algorithm.

→ Belady's Anomaly : Increasing frames can increase page faults.

Example:

Reference → 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

- 3 frames → 9 faults
- 4 frames → 10 faults

⑦ Explain the least Recently used (LRU) page replacement algorithm. How is it implemented?

→ LRU: Replace the page that was least recently used.

Implementation:
- counters (timestamps)
- Stack method

(8) What is Thrashing? What are its causes and how can the system detect it?

→ Thrashing: System spends more time swapping pages than executing.

Causes:

- Too many processes
- Insufficient frames

Detection:

- High page fault rate
- low CPU utilization

(9) Explain the working set model for thrashing prevention.

→ Working set = Pages used recently.

Idea:

- Allocate enough frames to hold working set.
- Prevents excessive page faults.

(10) Define a file. List and explain the various file attributes.

→ File : collection of related data stored on disk.

Attributes:

- Name
- Size
- Type
- Location
- Permissions
- Creation / modified time

(11) Describe the various file operations that an operating system Provides.

→
- Create
- Open
- Read
- Write
- Close
- Delete
- Seek

(12) Differentiate between sequential Access and Direct Access methods.

→ Sequential | Direct

- Access in order
- Slow for large files
- Simple

- Random access
- Fast
- Complex

(13) Explain the Tree-Structured Directory structure with a diagram. What are its advantages?

→ Structure: Hierarchical (like folders inside folders)

Advantages:

- Organized storage
- Easy Searching
- Supports grouping

(14) What is file system Mounting? why is it necessary?

→ Mounting: Attaching a file system to a directory.

Need:

- Access files from different devices.
- Integrate storage into one system

(15) Explain any two file allocation methods with their advantages and disadvantages.

→ 1. contiguous Allocation
- fast access
- External fragmentation

2. Linked Allocation
- No fragmentation
- Slow random access

(16) a) consider the following page reference string:
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for the following replacement algorithms assuming 3 and then 4 frames? Remember all frames are initially empty.

I) FIFO
II) optimal
III) LRU

⇒ Reference string:
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6

i) FIFO (first - in first out)
- 3 frames → 16 page faults
- 4 frames → 14 page faults

ii) optimal
- 3 frames → 11 page faults
- 4 frames → 8 page faults

iii) LRU (Least Recently used)
- 3 frames → 15 page faults
- 4 frames → 10 page faults

b) Based on your calculations, which algorithm performed best ? why?

→ It replaces the page that will not be used for the longest time in the future.

- This gives the minimum possible page faults.

(Algorithm)	(3 frames)	(4 frames)
FIFO	16	14
LRU	15	10
optimal	11	8

17) a) Explain in detail the concept of "Thrashing". Discuss the causes and the working-set model as a solution.

→ Thrashing: system spends more time paging than executing.

Causes:

- Too many processes
- Less memory
- High page faults

Working-set model:

- Keeps required pages in memory
- Allocates enough frames
- Reduces page faults

b) What is the "Page fault frequency" (PFF) approach to control thrashing?

→ PFF: controls thrashing using page fault rate.

- High fault rate → increases frames
- Low fault rate → decreases frames
- Maintains optimal performance

18. a) Discuss the various Directory structures (single-level, Two-level, Tree-structured, Acyclic-graph, General graph). Compare them in terms of file naming, sharing and complexity.

→
- Single-level: simple, no grouping, naming conflicts
- Two-level: seprate, user dirs, less conflict
- Tree: Hierarchical, efficient search
- Acyclic graph: Allows sharing, no cycles
- General graph: full sharing, cycles possible.

Comparison:

- Naming: Tree > Two-level > single
- Sharing: Graph > Acyclic > Tree
- Complexity: General graph (highest)

b) What are the challenges in implementing a general graph directory structure? How are they resolved?

→ Challenges:
- Cycles
- Infinite loops
- Deletion issues

Solutions:
- Garbage collection
- Reference counting

19.a) Describe ~~#~~ in detail the three main file allocation methods: contiguous, linked, and Indexed. Compare their performance for sequential and direct access.

→ • Contiguous: Fast, but fragmentation
- Linked: No fragmentation, slow random access
- Indexed: Balanced, supports direct access

Performance
- Sequential - contiguous best
- Direct - Indexed best

b) Explain how the UNIX inode implements indexed allocation using direct, single indirect, and double indirect blocks.

→ • Direct blocks: store file data directly
• Single indirect: Points to block of addresses
• Double indirect: Points to blocks of pointers
→ Supports large files efficiently.

20. a) What is the difference between a page and a frame? Explain the process of address translation in a paging system with a TLB.

→ Page: Logical memory unit
Frame: Physical memory unit

Translation:
• CPU → logical address
• TLB checks mapping
• If miss → page table lookup
• Get frame number.

R.V.U.

b) Given a 32-bit logical address space, a page size of 4KB, and a page table entry size of 4 bytes. Calculate the size of the page table. What problem does this create? How is it solved using multi-level paging?

→ • Page size = 4KB → offset = 12 bit
• Page = 2^{20}

Page table size:
= $2^{20} \times 4$ bytes
= 4 MB

Problem:
- Large memory usage

Solution:
- multi-level paging reduces size.

15.5.26